INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

GRINDING MACHINES, SURFACE, RECIPROCATING TABLE, HORIZONTAL SPINDLE, HAND FEED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers floor mounted, horizontal spindle, reciprocating table, hand feed grinding machines with equipment and accessories specified herein.

1.2 Classification. The grinding machine shall be of the following sizes. The size to be furnished shall be as specified (see 3.5 and 6.2).

Size	Worktable working surface, width (inches) x length (inches)
512	5 x 12
612	6 x 12
618	6 x 18
818	8 x 18

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3415

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

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(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 218.01 - Standard for Rating the Pitting Resistance and Bending Strength of Spur and Helical Involute Gear Teeth.

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Bevel, Hypoid, Fine Pitch Wormgearing and Racks only as Unassembled Gears.

AGMA 2000-A88 - Gear Classification and Inspection Handbook.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, Suite 201, 1500 King Street, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts, Tongues, and Cutters. (DOD adopted)

ANSI B7.1 - Safety Requirements for the Use, Care, and Protection of Abrasive Wheels. (DOD adopted)

ANSI B11.9 - Safety Requirements for the Construction, Care, and Use of Grinding Machines. (DOD adopted)

ANSI B74.2 - Specifications for Shapes and Sizes of Grinding Wheels, and Shapes, Sizes, and Identification of Mounted Wheels. (DOD adopted)

ANSI B74.13 - Markings for Identifying Grinding Wheels and Other Bonded Abrasives. (DOD adopted)

ANSI/ISO 54 - Cylindrical Gears For General Engineering and for Heavy Engineering- Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

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NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of precision surface grinding. The workpiece shall be supported on a reciprocating table and shall be ground using the periphery of a horizontally mounted grinding wheel. Table and grinding wheel feed shall be controlled manually. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, and repair.

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

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3.2.1.1 Dials. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position. All dials shall have independent zero adjustments except dual feed (coarse and fine feed) dials on which only the coarse feed dials shall have an independent zero.

3.2.1.2 Digital readout system. When specified (see 6.2), a digital readout system shall be installed on the Y axis (vertical head location) of travel. When specified (see 6.2), a digital readout system shall also be provided on the Z axis (cross feed position) of travel. The system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to ensure complete operation of the system. The system shall provide a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.00005 inch. The system accuracy shall be equal to the system resolution, verified upon installation and calibration of each system on the machine. The repeatability shall be 0.0001 inch, measured over the full travel on each axis. Special requirements for the system components shall be as specified (see 6.2). The unit shall indicate axis location in both inch-pound and metric units. An input power source fluctuation of $\pm$10 percent shall not adversely affect the digital readout system operation.

3.2.1.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type with a resolution equal to the resolution of the digital readout system. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination. Position feedback devices shall be easily and quickly interchangeable between like machines in any multiple machine order.

3.2.1.2.2 Electronic display console. The electronic display console for the DRO system shall contain a universal digital display with individual illuminated digits of not less than 0.43 inch in height. Digits shall be high efficiency displays for high light ambient conditions with wide viewing angle and shall be readable at a distance of not less than 16 feet. The console shall display at least six digits per axis simultaneously. The display console shall also provide keypad entry, memory, absolute/incremental counting, data

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preset, power failure indicator, inch/metric conversion, and absolute zero recall. The electronic display console shall be installed so as to be readily accessible and easily read from the operator's work station. The console shall be mounted to withstand shock and vibration generated by the machine. The connecting cable shall be armored and shall be installed so as not to interfere with the normal operations of the machine.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, remanufactured, or rebuilt products are allowed under this specification.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s). All controls shall be functionally grouped and clearly identified. An emergency control to stop all machine operations shall be provided.

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI B11.9. If a conflict arises between 29 CFR 1910 and ANSI B11.9, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices shall be provided for the point of operation and all other parts of the machine that present safety hazards.

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine or its components.

3.2.8 Environmental protection. During the operation, service, transportation, or storage of the machine, it shall not emit materials hazardous to the ecological system as prohibited by applicable Federal, state, or local statutes in effect on the date of the contract. Provisions should be made, as necessary, to provide for grinding dust collection during dry grinding (see 3.4.11.4).

3.2.9 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have at least a 24-hour capacity and means for determining fluid level. Means shall be provided to indicate a low oil condition. All oil holes, grease fittings, and filler caps shall be readily accessible.

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3.2.10 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts except for hand-fitted parts such as ways shall be interchangeable without requiring modification. Tooling and accessories normally supplied as standard items shall also be considered replacement parts. Replacement parts shall be domestically available.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the machine shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears of spindle and axis drive trains shall conform to or exceed the requirements of AGMA 218.01, AGMA 390.03, and AGMA 2000-A88 for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine.

3.3.8 Machine ways. Unless otherwise specified (see 6.2), all way surfaces shall be hardened, precision ground, and precision fit for smooth, accurate travel. Machine ways shall be protected regardless of machine component position to exclude contaminants.

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3.3.9 Slides. All sliding members shall be provided with adjustable gibs and guides, or other means, as necessary, to maintain the alignment and accuracy tolerances required herein for the machine type specified.

3.3.10 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist of not less than a base, saddle, worktable, wheelhead column(s), grinding wheelhead assembly, and feed mechanisms. Any additional components necessary for the operation of the machine shall be furnished in accordance with the manufacturer's standard practice.

3.4.1 Base. The base shall be a box-type, stress-relieved casting of iron or iron alloy, ribbed and braced for rigidity. The base shall support the saddle, worktable, column, and wheelhead assembly. Means shall be provided to level and secure the machine to a flat mounting surface. Longitudinal ways provided on the base shall be not less than one and one-half times the length of the worktable working surface. Hydraulic and coolant reservoirs, as applicable, may be housed in the base or provided as separate external units. Access shall be provided, as necessary, to service components housed in the base.

3.4.2 Saddle. The saddle shall be a casting of iron or iron alloy. The saddle shall traverse cross-axially (Z axis) either on guideways supported by the base or on guideways supported by the column. When the machine incorporates a saddle and worktable arrangement on the base, the top of the saddle shall have guideways not less than one and one-half times the length of the worktable working surface to support longitudinal worktable movement. When the machine incorporates a saddle and wheelhead assembly arrangement on the column, the saddle shall have guideways for facilitating cross-axial movement of the wheelhead assembly. Means shall be provided to prevent movement of the saddle after positioning during plunge or wheel form grinding. Stops shall be provided at each end of travel to prevent overtravel.

3.4.3 Worktable. The worktable shall be a casting of iron or iron alloy, ribbed to minimize distortion. The worktable shall travel longitudinally (X axis) on the guideways of the base or saddle, as applicable. The worktable shall be machined over the entire working surface. Means shall be provided to return spent coolant to its reservoir. Unless otherwise specified (see 6.2), the number of table T-slots shall be manufacturer's standard and shall meet the requirements of ANSI B5.1. Means shall be provided to prevent table overtravel. A T-slot or dog rack shall allow precise adjustment of reversing dogs. A grit deflector shall be provided. Means shall be provided for holding a magnetic chuck.

3.4.4 Electromagnetic chuck. When specified (see 6.2), the machine shall be furnished with an electromagnetic chuck and all accessories necessary for installation and operation, including necessary equipment to magnetize and demagnetize the chuck. The chuck shall have full and variable holding power.

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The chuck shall be sealed to prevent fluid leakage into the coils or accumulation of moisture therein. The poles of the chuck shall be not more than 1/8 inch apart over the entire face of the chuck to ensure small parts as well as large parts will be held securely during grinding. The chuck shall operate from the same power source as the machine and shall be completely wired and connected to the main power panel on the machine. When specified (see 6.2), a safety interlock shall be provided to prevent spindle rotation unless the chuck is energized. The size of the chuck working surface and other special chuck features shall be as specified (see 6.2).

3.4.5 Permanent magnetic chuck. When specified (see 6.2), the machine shall be furnished with a permanent magnetic chuck. The chuck shall be lever operated using no electric current. The size of the chuck working surface and other special features shall be as specified (see 6.2). When specified (see 6.2), a safety interlock shall be provided to prevent spindle rotation unless the chuck is energized.

3.4.6 Wheelhead column(s). The wheelhead assembly shall be mounted on a column(s). The column(s) shall be a one-piece, box-like casting, ribbed for rigidity and strength to minimize distortion, deflection, and vibration under normal workloads. The column(s) shall support and guide the wheelhead assembly or sliding saddle through its full range of vertical adjustment on vertical (Y axis) guideways provided on the column(s). When a sliding saddle arrangement is supplied, guideways shall be provided for transverse movement of the wheelhead. When a sliding column(s) is supplied, its slides shall fit appropriate guideways on the base or column bed. Column surfaces, when fitted to the base surfaces, shall provide accuracies required herein.

3.4.7 Grinding wheelhead assembly. The grinding wheelhead assembly shall include a housing, a grinding wheel spindle, and spindle drive mechanism. The assembly shall travel vertically on the wheelhead column. When the machine incorporates cross-axial wheelhead movement, either the wheelhead shall traverse cross-axially on a saddle on guideways provided on the column or the wheelhead column shall traverse cross-axially on base guideways. When specified (see 6.2), either the wheelhead shall tilt or sine plates shall be provided to allow tilt of the wheel on the part from horizontal to 30 degrees below the centerline about the center of the wheel.

3.4.7.1 Grinding wheelhead spindle and spindle drive. The spindle shall be of heat treated and ground steel alloy, supported and held in axial and radial alignment by precision, preloaded, antifriction ball or roller bearings. The spindle shall be dynamically balanced and provide vibration-free operation at all spindle speeds. Continuous or permanent lubrication to the spindle and bearings shall be provided. The spindle shall accept a driving mechanism at one end and a grinding wheel at the other end. The spindle shall accept taper-style wheel mounts or adapters for use with premounted and balanced grinding wheels. Spindle rotation shall be accomplished through a direct drive motor or belt and pulley arrangement. Motor speed shall be constant. When a belt and pulley arrangement is supplied, means shall be provided to adjust and maintain belt tension. All belts required for specified wheel speeds shall be furnished.

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3.4.7.2 Grinding wheels. Grinding wheels shall conform to ANSI B74.2. The wheels shall be marked in accordance with ANSI B74.13 and shall meet the safety requirements established in ANSI B7.1. All grinding wheels shall be properly guarded. When specified (see 6.2), wheel guards shall have means to connect a dust extraction unit (see 3.4.11.4).

3.4.7.3 Grinding wheelhead spindle drive. Spindle rotation shall be accomplished through a direct drive motor or a motor and belt and pulley arrangement. The spindle drive shall provide operating speed(s) in accordance with manufacturer's standard and safe operating speeds of wheels recommended, but not less than 4500 surface feet per minute (SFPM) on minimum safely usable wheel diameter. Motor speed shall be constant. When a belt and pulley arrangement is supplied, means shall be provided to adjust and maintain belt tension. All belts required for specified wheel speeds shall be furnished.

3.4.8 Wheel truing device. Unless otherwise specified (see 6.2), the machine shall have a permanent diamond truing device arranged to mount on the worktable or magnetic chuck, when applicable. When specified (see 6.2), each machine shall be equipped with an in-place diamond truing device arranged to in-feed and to dress the entire periphery of the grinding wheel in lieu of a work surface mounting diamond. The diamond shall be not less than one carat. The truing device shall permit wheel dressing without disturbing the workpiece set-up or changing the wheelhead vertical setting prior to wheel dressing. The device shall true wheels up to and including the size given in table I for the machine size specified.

3.4.9 Feed mechanisms. The machine shall be provided with manual means for controlling worktable traverse, crossfeed, and the vertical movement of the grinding wheelhead assembly. The crossfeed (Z axis) and vertical (Y axis) feed controls shall be graduated in increments of 0.001 inch or less, directly readable on a dial or vernier. When specified (see 6.2), fine feed dials shall also be provided for crossfeed and down feed, graduated in increments of not greater than 0.0001 inch.

3.4.10 Coolant system. Unless otherwise specified (see 6.2), a coolant system shall be provided. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining, cleaning, and determining the fluid level. Means shall be provided to permit the operator to direct and control the amount of coolant. An interlock shall be provided to prevent the use of a dust collection system (see 3.4.11.4) during wet grinding. Means shall be provided to select either wet grinding (coolant) or dry grinding with dust extraction.

3.4.11 Electrical system. The electrical system shall be wired to conform to NFPA 79. Unless otherwise specified (see 6.2), each machine shall be wired to draw all of its electrical power from a single 230/460-volt, 3-phase, 60-Hertz (Hz) circuit, with a maximum line voltage fluctuation of 10 percent rated voltage. Unless otherwise specified (see 6.2), the machine shall be initially wired for operation on 230 volts.

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3.4.11.1 Motors. Motors shall meet the requirements of 3.4.11. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Each motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.11.2 Worklight. When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield and an on-off switch and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.11.3 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.4.11.4 Dust extractor. Unless otherwise specified (see 6.2), a dust extractor and all necessary connections shall be provided to be used during dry grinding (an individual dust extraction unit is not necessary if a central dust extraction system is available). The dust extractor shall be connected to the machine's electrical system with start/stop pushbuttons conveniently located on the control panel. The dust extractor shall have the special features and be of the efficiency rating as specified (see 6.2). An interlock shall be provided to prevent wet grinding (see 3.4.10) while using the dust extractor. Means shall be provided to select either wet grinding (coolant) or dry grinding with dust extraction.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the machine shall be not less than specified in table I for the machine size specified.

3.6 Alignment tolerances. Each machine shall meet the alignment tolerances specified in table II.

3.7 Performance. The machine shall meet the performance requirements given in 4.7.4.

3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment. Unless otherwise specified (see 6.2), the following equipment shall be furnished as standard with each machine:

a. One set of special wrenches, operating, and repair tools normally furnished by the manufacturer.

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b. One set of straight grinding wheels normally furnished by the manufacturer.

c. One extra set of wheelhead belts, if applicable.

3.9 Optional equipment. The following equipment and accessories are commercially available and are considered optional. When required, optional equipment, quantity and description, shall be as specified (see 6.2).

a. Additional grinding wheels: shank size, number of each wheel size, type, dimensional characteristics, and composition of each wheel, as specified (see 6.2).

b. Other optional equipment as specified (see 6.2).

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.11 Marking on instruments, control panels, and plates. All words on instruments, control panels, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate. When specified (see 6.2), a lubrication plate or chart shall be attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder or permanently sealed between clear plastic sheets. The information provided on the chart or plate shall be as specified by the procuring activity (see 6.2).

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall provide the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Workmanship. Workmanship of the machine and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except

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as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The machine shall pass the first article examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. The machine shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Test conditions. All inspections, tests, and examinations shall be performed in an indoor facility with ambient conditions of 41° to 104° Fahrenheit and 20 percent to 95 percent relative humidity.

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4.7.2 Operational test. The machine shall be operated at no load for not less than 1 hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment tolerance test shall be performed prior to and subsequent to the performance test(s) given in 4.7.4. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.7.4 Performance tests. The machine shall be subjected to the following performance test. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.7.4.1 Block grinding test. The machine shall be tested to determine its ability to surface grind. The machine shall grind the top and bottom surfaces of five test blocks, one located at each corner and one in the center of the worktable. The test blocks shall be the manufacturer's standard test blocks. The blocks shall have a 4-square inch surface area and be 1-inch thick (minimum). The method and grinding wheel(s) used to perform this test shall be at the option of and supplied by the manufacturer. The machine shall surface grind the top and bottom of the five test blocks at any location on the table working surface parallel to within 0.0001 inch per foot and flat to within 0.0001 inch per foot. The five test blocks shall be ground equal in size to within 0.0001 inch per foot. The last surface ground shall have a finish not rougher than 8 microinch.

4.7.4.2 Additional performance tests. Performance tests required in addition to or in lieu of the requirements given in 4.7.4.1 shall be as specified and fully described (see 6.2).

4.7.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.10 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The machines covered by this specification are for use in tool and die, maintenance, and production shops for rough and precision finish grinding, slot, or form grinding operations to be performed on a wide variety of materials.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (see 3.2.1).

f. Digital readout system, if required (see 3.2.1.2).

g. Specify digital readout system on Z axis (cross feed position), if required (see 3.2.1.2).

h. Specify any special digital readout system requirements, if required (see 3.2.1.2).

i. Additional safety and health requirements, if required (see 3.2.5).

j. Painting, if different (see 3.3.5).

k. Way surfaces, if different (see 3.3.8).

l. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.10).

m. Table T-slots, if different (see 3.4.3).

n. Electromagnetic chuck, if required (see 3.4.4).

o. Safety interlock on electromagnetic chuck, if required (see 3.4.4).

p. Electromagnetic chuck working surface size and special features, if required (see 3.4.4).

q. Permanent magnetic chuck, if required (see 3.4.5).

r. Permanent magnetic chuck working surface size and special features, if required (see 3.4.5).

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s. Safety interlock on permanent magnetic chuck, if required (see 3.4.5).

t. Wheelhead tilt, if required (see 3.4.7).

u. Wheel guards with dust extractor connections, if required (see 3.4.7.2).

v. Mountable truing device, if not required (see 3.4.8).

w. In-place truing device in lieu of a mounted truing device, if required (see 3.4.8).

x. Fine feed dial on crossfeed and down feed, if required (see 3.4.9).

y. Coolant system, if not required (see 3.4.10).

z. Electrical system, if different (see 3.4.11).

aa. Initial input voltage, if different (see 3.4.11).

bb. Worklight, if required (see 3.4.11.2).

cc. Hour meter, if required (see 3.4.11.3).

dd. Dust extractor, if not required (see 3.4.11.4).

ee. Special dust extractor features and efficiency rating, if required (see 3.4.11.4).

ff. Size and capacity, if different (see 3.5).

gg. Additional performance features, if required (see 3.7.1).

hh. Standard equipment, if different (see 3.8).

ii. Specify and fully describe optional equipment, if required (see 3.9).

jj. Specify additional grinding wheels (shank size, number of each wheel size, type, dimensional characteristics, and composition of each wheel), if required (see 3.9.a).

kk. Specify applicable repair parts, if required (see 3.10).

ll. Lubrication plate, if required (see 3.11.1).

mm. Specify information to be provided on lubrication plate (see 3.11.1).

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nn. Nameplate, if different (see 3.11.2).

oo. Inspection conditions, if different (see 4.3).

pp. First article inspection, if different (see 4.4).

qq. Quality conformance inspection, if different (see 4.5).

rr. Additional performance tests, if required (see 4.7.4.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
.	.	.	.
.	.	.	.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD From 1423.

6.3.1 Technical data. All technical data ordered using a DD Form 1423 should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

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6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification changes. Classification changes from the previous issue of this specification are as follows:

MIL-G-80145A	MIL-G-80145B
Size 510	Deleted
New	Size 512
Size 612	Size 612
New	Size 618
Size 815	Size 818
Size 1015	Deleted

6.8 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

6.11 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.13 Subject term (keyword) listing.

Abrasion
Grinding
Grinding wheels
Machine tools
Magnetic chuck
Manual

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TABLE I. Sizes and capacities.

Characteristic	Size 512	Size 612	Size 618	Size 818
Capacity:				
Worktable surface, inches	5 x 12	6 x 12	6 x 18	8 x 18
Table top to bottom of new standard wheel, inches	12	12	11-3/4	15
Cross feed travel, inches	5-1/2	6-3/4	6-3/4	8-3/4
Longitudinal table travel, inches	12	12-3/4	19	22
Vertical travel, inches	12	12-1/2	12-1/2	14-1/2
Standard grinding wheel size (new), diameter x thickness, inches	7 x 1/2 x 1-1/4			
Feeds and speeds:				
Spindle speed, RPM	3450			3600
Table longitudinal feed	Manual			
Cross feed positioning	Manual			
Vertical positioning	Manual			
Motor horsepowers (minimum):				
Spindle	1			1-1/2

NOTE: Machine capacities shall be not less than the stated requirements of the size ordered. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater. Physical size limitations of width, length, or height shall be as specified by the procuring activity to ensure the machine will fit into its future operating location (see 3.5).

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6.14 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specifications are created as follows:

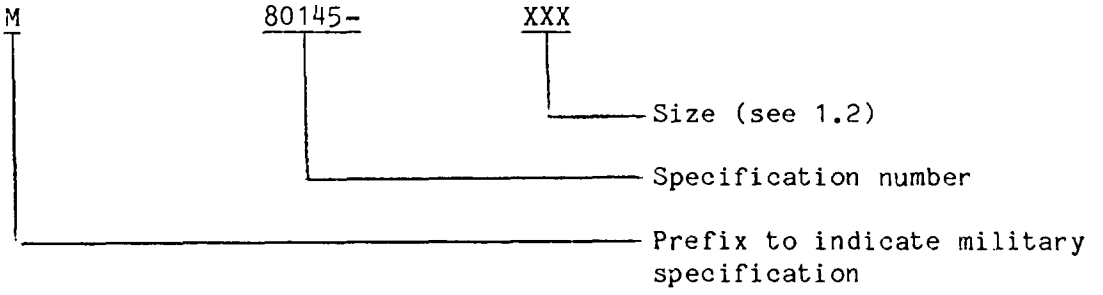


TABLE II. Alignment and accuracy requirements.

Test to be applied	Permissible tolerance
Worktable flat lengthwise, crosswise, and diagonally (no twist permitted)	0.0002"/FT
Rise and fall of worktable in its longitudinal movement	0.0001"/FT
**Worktable T-slots parallel with worktable in longitudinal movement	0.0005"/FT
Squareness of longitudinal and cross feed travel	0.001"/FT
Periphery runout of wheel spindle	0.0002" TIR
Wheel spindle axial slip	0.0001" TIR
Wheel spindle axis square to worktable surface (using a 6", 180 degree radius sweep in a vertical plane)	0.0005" TIR
Wheel spindle axis parallel to worktable surface	0.0007"/18"
Wheel spindle axis square with worktable longitudinal travel	0.0004"/FT
Vertical adjustment of wheelhead assembly square with worktable surface:	
- in a longitudinal plane	0.003"/FT
- in a transverse plane	0.003"/FT

** When applicable

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - SM
Air Force - 84
DLA - GS

User activities:
Army - SM, AR
Navy - MC, SH

Preparing activity:
DLA - IP

Project (3415-0171)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-80145B	Grinding Machines, Surface, Recip Table, Horiz Spindle, Hand

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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